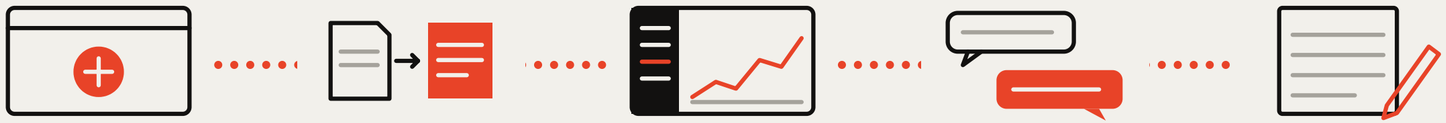
KaxaNuk
Sharing knowledge

— INVESTMENT BOOTCAMP

# READY FOR PAPER TRADING? VALIDATE, FREEZE AND PRESENT.



**06**

From S05's finding to the decision: nothing wrong, the five criteria, frozen to watch it out of sample, and presented to a committee.

IN THIS SESSION


01 VALIDATE

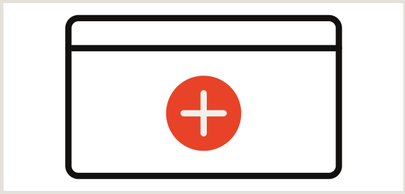
02 DECIDE

03 FREEZE

04 PRESENT

05 WRAP-UP

# FROM YOUR FINDING TO YOUR DECISION.

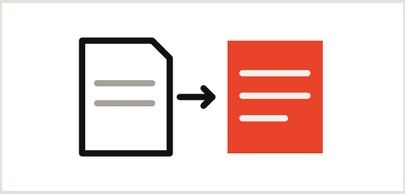


## 1 VALIDATE

That nothing is wrong.

- What cannot fail
- The blueprint judges the run
- The holdout, once

You leave with: your validation

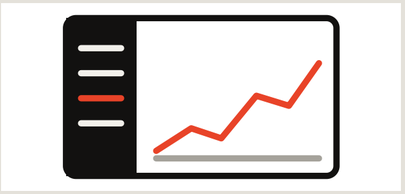


## 2 DECIDE

Five criteria. All of them.

- The gate, criterion by criterion
- Your book's verdict
- The signature is a person's

You leave with: your verdict

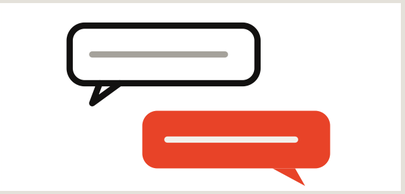


## 3 FREEZE

To watch it out of sample.

- The book, frozen as it is
- A test day first
- Every day, after the close

You leave with: your frozen book



## 4 PRESENT

To a committee.

- The rubric's nine rows
- Answer with a file
- Ten minutes

You leave with: your presentation



## 5 WRAP-UP

What you learned, and what comes next.

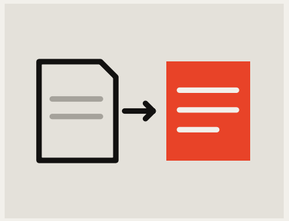
- The session's ideas
- Homework: close part H
- Next: the pillars

You leave with: your next step

# EIGHT STEPS. IN THIS SESSION, THE SEVENTH.

S02 TO S05 | THIS SESSION | BEYOND THE COURSE

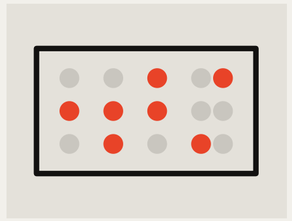
1 2 3 4 5 6 7 8



## RESEARCH

S02

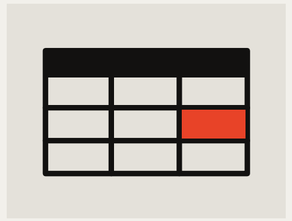
Your objective, from your library · OBJECTIVE.md



## UNIVERSE

S02

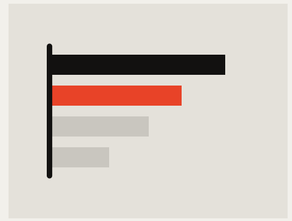
Which companies and when; delisted included · Universe/



## DATA

S03

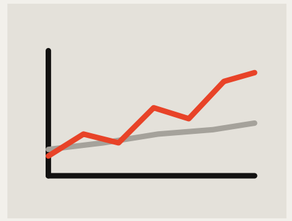
Your features, and whether they carry a signal · Data/



## PORTFOLIO

S04

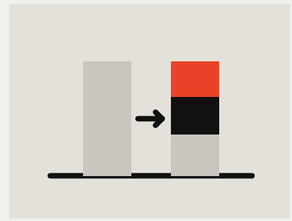
Your hypothesis, then the portfolios · Experiments/



## BACKTEST

S05

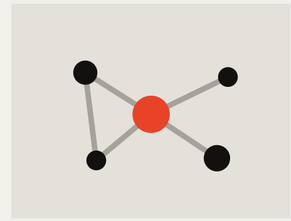
Your portfolio through time, net of costs



## ATTRIBUTION

S05

Where the return came from



## PAPER TRADING

S06

Tested without real money · Paper_Trading/

STEP 7



## PRODUCTION

outside the course

**From S05's homework**

your `FINDINGS_1.md`, with your verdict, and N.

# BEFORE DECIDING: WHAT CANNOT FAIL.

| THE CHECK | WHERE IT SHOWS | FROM |
|---|---|---|
| **The blueprint, saved first** | Its version's time, before the rule. | S04 |
| **What was known that day** | The signal at t – 1; the history before the date. | S04 |
| **The dead, included** | The point-in-time seed; the coverage. | S02 · S03 |
| **Net, and every day priced** | The complete run, with its two cost rows. | S05 |
| **N counted** | Every signal and every book, in the journal. | S03 to S05 |
| **The holdout, once** | Its prediction, saved before opening it. | S06 |
| **It reproduces** | From a clean clone, with one command. | S06 |

**IF ONE FAILS** you do not decide: you fix it, and it counts.

# THE BLUEPRINT JUDGES THE RUN.

ARYA · TREND-AND-VOLUME

`challenge`: my run against my blueprint.
1

Each prediction against its falsifier, with its file. Go ahead?

go ahead

Prediction 1 held. The benchmark did not.
2

THE PREDICTION, AGAINST THE RESULT · NET

| NET SHARPE | VALUE | |
|---|---|---|
| The rule, with the 50/200 cross | 0.30 | |
| Its control, without the cross | 0.25 | |
| Margin over the control | +0.05 | 3 |
| KN US Equity Core | 0.41 | |

Prediction 1: more than +0.03 over its control. It held.

1. Ask for the challenge; read each line.
2. Each prediction against its falsifier.
3. The cross adds; the book does not beat the benchmark.

`challenge`

# A WINNER MEANS LITTLE WITHOUT ITS N.

1999–2019

CORRECT FOR THE SEARCH

1999
**SULLIVAN, TIMMERMANN & WHITE**

THE UNIVERSE OF RULES
A trading rule, judged against all it was drawn from.

2000
**WHITE**

A REALITY CHECK
Does a search's best model beat its benchmark?

2015
**HARVEY & LIU**

THE HAIRCUT
A backtest's Sharpe, cut for the tests behind it.

WHAT THE SEARCH COSTS

2016
**WIECKI, CAMPBELL, LENT & STAUTH**

OUT OF SAMPLE
888 algorithms: the backtest Sharpe predicts little.

2017
**SUHONEN, LENNKH & PEREZ**

BACKTEST TO LIVE
215 strategies: Sharpe ratios fell once live.

2019
**LÓPEZ DE PRADO**

FALSE DISCOVERIES
Most are false: selection bias under many tests.

1

2014
**BAILEY & LÓPEZ DE PRADO**

THE DEFLATED SHARPE
Corrects for many tests and non-normal returns.

WHAT CHANGED

**The best of many trials is the largest of many draws.**

THEREFORE

Publish N. Then deflate the winner, with ①.

Each line from its paper's published abstract.

# OPENED ONCE, AFTER ITS PREDICTION.

| NET | THE RULE | ITS CONTROL | KN US EQUITY CORE |
|---|---|---|---|
| **In sample, 2000 to 2024** | 0.24 | 0.20 | 0.38 |
| **Holdout, 2025 to 2026** | 1.04 | 0.98 | 0.93 |
| **Annual return, in sample** | 4.6% | 3.4% | 7.7% |
| **Annual return, holdout** | 40.0% | 35.0% | 20.5% |
| **Max drawdown, holdout** | −28.3% | −27.3% | −19.4% |

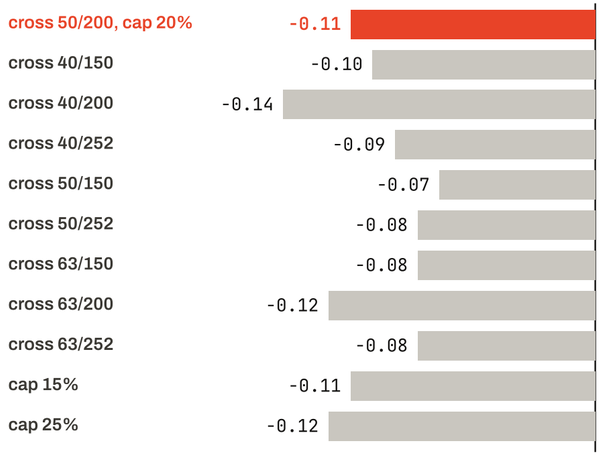
**A YEAR AND A HALF** little evidence: it confirms, it does not decide.

# DOES IT HOLD IF YOU MOVE IT A LITTLE?

NET SHARPE, LESS THE KN US EQUITY CORE'S · N = 11

Sharadar · KN US Equity Core · provisional

EACH VARIANT, THE SAME RULE; THE CROSS OR THE CAP CHANGES




Net of 10 basis points per side, 2000 to 2026. No variant beats the benchmark: the curve is flat, and below zero.

**A curve, not a peak**

flat: the conclusion does not depend on 50/200.

# FIVE CRITERIA. ALL OF THEM.

| CRITERION | EVIDENCED BY |
|---|---|
| **1 · Beats its benchmarks and its control, risk-adjusted** | FINDINGS_1.md's table, on the same window. |
| **2 · Idiosyncratic return in both layers** | Brinson-Fachler and the factor model. |
| **3 · Robust, with N published** | A curve of variants; the trial count. |
| **4 · Costs and capacity, modelled and stated** | The cost rows; the capacity. |
| **5 · A signature** | A name and a date in BITACORA.md. |

**IF IT IS NOT EVIDENCED** by FINDINGS_1.md and RESULTS.md, it is not met.

# YOUR BOOK'S VERDICT.

| CRITERION | THE EVIDENCE | |
|---|---|---|
| **1 · Benchmarks** | Sharpe 0.30 against the KN US Equity Core's 0.41; it beats its control. | No |
| **2 · Idiosyncratic** | Selection of +1%; Jensen's alpha, the intercept, negative. | No |
| **3 · Robust** | Eleven variants, a flat curve; N published. | Yes |
| **4 · Costs** | Costs modelled; capacity, not. | No |
| **5 · Signature** | Not sought: 1, 2 and 4 block. | — |

**IT DOES NOT PASS** the blocking rows are the content: on to experiment 2.

# REJECTED, WITH THE SAME CLARITY.

PAPER_TRADING/BITACORA.MD · CURRENT STATUS

```
## Current status
Experiment 1 · the 50/200 rule: not graduated.                    1
Blocks: 1 (below the KN US Equity Core), 2 (Jensen's alpha, the intercept,
  negative), 4 (capacity not modelled).
Clears: 3 (eleven variants, a flat curve; N published).
Criterion 5: not sought.
Next: Experiment 2, a new blueprint.                              2
```

1. A no, with the same shape as a yes.
2. A new idea is another experiment.

**What failed, too**

stays in the record: it is what you learned.

# THE SIGNATURE IS A PERSON'S.

**Criterion 5**

A name and a date.

**Your researcher**

Lays out 1 to 4, evidenced or not, and stops.

**It never declares**

That the strategy graduates.

**It never computes**

A figure the files do not hold.

**You sign**

Or not, with the blocking rows.

The signature is sought when 1 to 4 are evidenced, never before.

# FREEZE THE BOOK, AS IT IS.

PAPER_TRADING/PAPER_TRADING_1/

| | |
|---|---|
| `paper_trading_1.py` | the rule, from the experiment's cells [1] |
| `FREEZE.json` | the commit, the date, each file with its hash, the versions [2] |
| `the refinery and its calculations` | copied, byte for byte |
| `the seed and the modules` | copied from the commit |
| `the security master` | from disk; git does not keep it |

**A changed file** stops the book: what is frozen does not move.

**Version 1.0.0:** the first book on paper trading, reproduced from a clean clone.

1. Promoted, not copied: the same number.
2. promote.py, on a clean tree.

**Never twice**

a frozen book is never frozen again: another freeze is another book.

# A TEST DAY BEFORE THE FREEZE.

## In a scratch clone

Frozen with no commit; one day with `--dry-run`.

### The membership

No members after the last file: it sells everything that day.

### The seed

A span closed on its last day: it stops the live ones.

### The new days

They are the first any code reads.

Both faults only show there: fix them before the real freeze.

# EVERY DAY, AFTER THE CLOSE.

DAILY_UPDATE.PY · ONCE A DAY

```
the day's prices, once · checked                                  1
each book: its frozen refinery and rule
the engine three times: the history, after the window, since the freeze
the record: six tables; a restatement is flagged, never overwritten   2
it exits 0, 1 or 2: clean, flagged, failed                        3
a day with no price or an impossible move: it stops
the scheduled task, you register it
```

1. A changed file: `unfrozen-input`.
2. The past is not rewritten.
3. Read the flags, not only the exit code.

# WHAT YOU WATCH OUT OF SAMPLE.

| WATCH | WHY |
|---|---|
| **Whether it behaves like its backtest** | Turnover, holdings, exposure, costs: within their bands. |
| **The kill switch** | Written first: what would stop it. |
| **The review dates** | When it is read, and how long until production. |
| **Nothing is refitted** | A run that tunes anything is another backtest. |

**A FEW MONTHS DO NOT PROVE SKILL** they prove the plumbing works.

# TWO TRACKS, ONE FROZEN BOOK.

## INVESTMENT LAB

NO CODE

Paper trading: your book →

its bands and its kill switch →

the record, every day

**Frozen:** the rule and its versions.

**Every day:** after the close.

**Bands:** from FINDINGS_1.md.

**Kill switch:** BITACORA.md's.

## ARYA · TREND-AND-VOLUME

Signed: freeze book 1. 1

First a test day in a clone. Then promote.py and FREEZE.json. Go ahead?

go ahead

Frozen. The command to schedule it is yours: here it is. 2

1. Only after the signature.
2. You register the scheduled task.

```
paper-trading-gate
```

# NINE ROWS. NONE FOR RETURN.

PRESENTATION DAY · A JUDGE'S SHEET

The 3x3 rubric · a row per file

| AREA | ROW | A JUDGE OPENS | 0 · 33 · 66 · 100 |
|---|---|---|---|
| Strategy definition | Hypothesis and source of returns | OBJECTIVE.md | |
| | Research foundation | Bibliotheca/ | |
| | Scientific process | git log | |
| Feature engineering | Data quality and integrity | Universe/ | |
| | Feature design and relevance | RESULTS.md | |
| | Signal construction | c_* · r_* | |
| Strategy design | Portfolio construction logic | BLUEPRINT_1.md | |
| | Backtesting and robustness | FINDINGS_1.md | |
| | Attribution and understanding | FINDINGS_1.md | |

reproduces | blueprint saved first | figures traced | N published | holdout once | gate evidenced

Six process gates beside the rows: pass or fail.

Equal weight: 0 not there · 33 basic · 66 clear · 100 rigorous, reproducible, documented. The handbook's reference pages: *Six questions, each to its row*; *Ten minutes, built from files*.

**A LOSS CAN SCORE 100** when it is measured, counted and explained.

# ANSWER A JUDGE WITH A FILE.

## BREAKOUT ROOM · GOLDEN-FLOW FIRST

**JUDGE · PORTFOLIO CONSTRUCTION LOGIC**
What is your control?

`BLUEPRINT_1.md`, *Rules.*

**JUDGE · BACKTESTING AND ROBUSTNESS**
How many did you try?

`RESULTS.md`, *The trial count.*

## YOUR PARTNER'S CARD

| 0 · 33 · 66 · 100 | A JUDGE ASKS | FILE · SECTION |
|---|---|---|
| | What is your control? | |
| | How many did you try? | |
| | Which factor is it? | |
| | Net of what? | |
| | Who is missing? | |
| | What would kill it? | |

Questions 1 to 3 on Golden-Flow, then swap for 4 to 6; then one each on your own repository. Share your screen; push nothing. Alone: your researcher asks, one question at a time; you answer with a file.

Each answer, a file and its section. Golden-Flow · RESULTS.md · FINDINGS_1.md · run 2026-10-06 · Backtest Engine 0.66.0 · Attribution Analysis 0.2.0 · hypothetical

# TEN MINUTES BEFORE THE COMMITTEE.

| MINUTE | WHAT YOU PRESENT | THE FILE |
|---|---|---|
| **0 to 2** | The hypothesis and why it should work. | OBJECTIVE.md |
| **2 to 3** | The universe and the benchmark, with the dead. | Universe/ |
| **3 to 5** | The rule and its control, saved first. | BLUEPRINT_1.md |
| **5 to 7** | Net, against the control and the benchmark; N. | FINDINGS_1.md |
| **7 to 8** | Where it came from: factors and sectors. | FINDINGS_1.md |
| **8 to 10** | The verdict, and what comes next. | BITACORA.md |

**A NO IS PRESENTED TOO** measured, counted and explained.

# THE SESSION'S IDEAS, IN YOUR WORDS.

**01**

What do you check before deciding?

**02**

Why the five criteria, all of them?

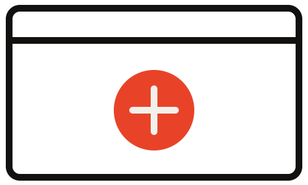
**03**

What is frozen, and why never twice?

**04**

What do you watch out of sample, and what not?

# BEFORE THE COMMITTEE: CLOSE PART H.



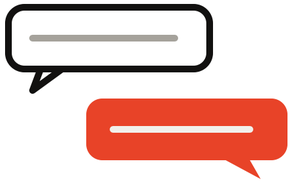
## 01 VALIDATE

- The list, check by check.
- Your holdout, once.
- Your curve of variants, with N.



## 02 DECIDE

- The five criteria, with their evidence.
- BITACORA.md: yes or no.
- If yes: the test day and the freeze.



## 03 PRESENT

- Ten minutes, with your files.
- Rehearse the six questions.
- Ask `next`.

# VALIDATED FIRST. FROZEN NEXT.

Your strategy, validated, decided in writing and, if it passes, frozen to watch it out of sample.

**LEARN**

The pillars go deeper into each step.

`github.com/KaxaNuk/Investment-Bootcamp`

**TEST**

Your strategy on the Lab; universes and factors from Analytics.

`kaxanuk.mx/lab` · `kaxanuk.mx/analytics`

**TELL US**

What your committee decided.

**QUESTIONS** `workshops@kaxanuk.mx`

NEXT

## The pillars: Coding Foundations, Financial Markets, Mathematical Finance

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Nothing here is a recommendation to buy, sell or hold any security; backtests are hypothetical, in sample and never traded; this session's figures are our own calculation on past data, not a result; paper trading uses no real money. Your researcher can be wrong; every decision is yours. What your researcher reads goes to the company behind your assistant, never to KaxaNuk.

Tools read 2026-10-13; works and links, 2026-10-11; the figures, from Sharadar prices of the KN US Equity Core, whose methodology is still provisional, computed 2026-10-17, with 10 basis points per side. Shared under CC BY 4.0; the KaxaNuk name, logo and isotipo are not.